HOSPITAL OPERATIONS & PATIENT ANALYTICS DASHBOARD

FINAL PROJECT DOCUMENTATION

Submitted as part of the Infosys Springboard Internship Project

Prepared By	SRIRAMOJI THARUN
Programme	Bachelor of Technology in Artificial Intelligence
Institution	Amrita Vishwa Vidyapeetham, Amaravati
Internship	Infosys Springboard Internship
Project Title	MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)
Project Duration	July 2026 – August 2026
Submission Year	2026

MedTrack DV — Hospital Operations & Patient Analytics Dashboard

An interactive healthcare analytics solution that turns raw hospital operational and patient data into actionable insights across hospital performance, patient flow, department efficiency, and resource utilization.

Live Repository: Hospital-Operations-Patient-Analytics-

Table of Contents

---

1. Executive Summary

MedTrack DV is an end-to-end hospital operations and patient analytics solution built to help hospital administrators, department heads, and operations managers monitor performance across admissions, patient flow, department efficiency, and resource capacity. Raw hospital data was collected, cleaned, and transformed using Python (Pandas, NumPy) and Power Query, then modeled and visualized in **Power BI** across four interconnected dashboards — **Hospital Overview, Patient Flow, Department Analytics,** and **Resource Utilization**.

The solution tracks core operational KPIs — including Total Admissions, Total Revenue, Average Length of Stay (LOS), Readmission Rate, Occupancy Rate, and Bed/Staff Utilization — and allows stakeholders to slice this data by Admission Date, Hospital Name, Department, Gender, and State. The result is a single, interactive analytics layer that supports faster, data-driven decisions on patient flow, capacity planning, and department performance.

> **Note on platform:** The original project specification referenced Tableau as the intended visualization tool. The actual implementation shown in this repository — and documented below — was built and delivered in **Power BI**.

---

2. Project Overview

Hospitals generate large volumes of operational and clinical data every day — admissions, discharges, department activity, bed occupancy, staffing, and revenue. Without a centralized analytics layer, this data remains siloed and difficult to act on in real time.

MedTrack DV consolidates this data into a governed, KPI-driven Power BI data model and presents it through four purpose-built dashboards, each targeting a specific operational lens: high-level hospital performance, patient movement through the system, department-level efficiency, and physical/staffing resource utilization.

3. Problem Statement

Hospital administrators and department heads often lack a consolidated, real-time view of operational performance. Metrics such as admissions, occupancy, length of stay, and readmissions are typically scattered across disconnected reports, making it difficult to:

- Identify departments with high patient volume or inefficient LOS.
- Track occupancy and bed availability to plan capacity ahead of demand.
- Monitor readmission trends that may signal care-quality issues.
- Compare revenue and performance across hospitals and departments.

MedTrack DV addresses this by unifying hospital, patient, department, and resource data into a single interactive analytics platform.

4. Project Objectives

- Build a centralized, interactive dashboard suite covering hospital, patient, department, and resource-level analytics.
- Engineer standardized KPIs (Admissions, Revenue, LOS, Readmission Rate, Occupancy Rate, Bed/Staff Utilization) from raw operational data.
- Enable multi-dimensional filtering (date, hospital, department, gender, state) for drill-down analysis.
- Surface trends in admissions, discharges, and revenue over time.
- Support capacity and resource-planning decisions through bed, ICU, and staff utilization views.
- Deliver a portfolio-ready, well-documented BI project suitable for GitHub and interview discussion.

5. Business Requirements

#	Requirement	Addressed By
1	Track hospital-wide admissions, revenue, and occupancy trends	Hospital Overview Dashboard
2	Monitor patient movement (admissions, discharges, admission type, gender)	Patient Flow Dashboard
3	Evaluate department-level volume, efficiency, and revenue	Department Analytics Dashboard
4	Assess bed, ICU, and staffing capacity	Resource Utilization Dashboard
5	Allow filtering by date range, hospital, department, gender, and state	Global filter panel (all dashboards)
6	Present KPIs in a consistent, comparable format across all views	Standardized KPI card layout

6. Target Users / Stakeholders

- **Hospital Administrators** — overall performance, revenue, and occupancy oversight.
- **Operations Managers** — patient flow, admissions/discharges, and capacity planning.
- **Department Heads** — department-specific volume, LOS, revenue, and efficiency.
- **Resource / Facilities Managers** — bed, ICU, and staff allocation across departments.
- **Data Analytics / BI Reviewers** — for evaluating the project's analytical and technical approach.

7. Dataset Description

The dataset represents hospital operational and patient-level records, including admission and discharge activity, hospital and department identifiers, patient demographics, and resource-related fields. Based on the fields visualized across the four dashboards, the dataset includes attributes such as:

Field Category	Example Fields Observed in Dashboards
Admission Details	Admission Date, Admission Type (Urgent / Emergency / Elective), Discharge Date/Trend
Hospital Details	Hospital Name, Hospital Type (Private / Government), State
Department Details	Department (Cardiology, Neurology, Oncology, General Medicine/Surgery, Psychiatry, Nephrology, ICU, Pulmonology, Orthopedics, etc.)
Patient Details	Gender, Length of Stay, Readmission status
Financial Details	Revenue (per hospital / per department)
Resource Details	Available Beds, Occupied Beds, ICU Beds, Staff Count

Note: Exact source column names, data types, and row-level schema are **To be confirmed**, as the underlying raw dataset file was not provided for direct inspection — the fields above are inferred from the KPIs and visuals present in the implemented dashboards.

8. Data Collection

Hospital operational data was collected as the foundation for the analytics workflow. Details on the original data source (e.g., synthetic dataset, public healthcare dataset, or simulated hospital records) are **To be confirmed**. The data was ingested into the project's Python/Power Query pipeline for cleaning and transformation prior to modeling in Power BI.

9. Data Cleaning and Transformation

Using **Python (Pandas, NumPy)** and **Power Query**, the raw hospital dataset was cleaned and standardized to support reliable KPI calculation and visualization. This included:

- Handling missing or inconsistent values across admission, department, and resource fields.
- Standardizing categorical fields (e.g., Department names, Hospital Type, Admission Type, Gender) for consistent grouping and filtering.
- Converting and validating date fields (Admission Date, Discharge-related fields) for accurate time-series trending.
- Deriving calculated fields required for KPI computation (e.g., Length of Stay, Occupancy, Readmission flags).

Specific transformation logic and exact cleaning steps beyond what is inferable from the dashboards are **To be confirmed**.

10. Data Preparation Workflow

```
flowchart LR
    A[Raw Hospital Data] --> B[Data Collection]
    B --> C[Data Cleaning<br/>Python: Pandas / NumPy]
    C --> D[Data Transformation<br/>Power Query]
    D --> E[KPI Engineering]
    E --> F[Power BI Data Model]
    F --> G[Dashboard Development]
```

11. KPI Engineering

Core operational KPIs were engineered from the cleaned dataset to be consistently reused across all four dashboards as standardized summary cards. This ensures that metrics like Total Admissions, Total Revenue, and Occupancy Rate mean the same thing regardless of which dashboard or filter context they appear in.

Where DAX was used to calculate these KPIs within the Power BI data model, exact formula logic is **To be confirmed**, as it was not directly inspectable from the screenshots provided.

12. KPI Definitions

KPI	Definition	Value Observed
Total Admissions	Total count of patient admissions across the selected filter context	10K
Total Revenue	Aggregate revenue generated across hospitals/departments	₹2.50bn
Average Length of Stay (LOS)	Average number of days a patient stays admitted	8.07 days
Readmission Rate	Percentage of patients readmitted within the tracked period	49.87%

KPI	Definition	Value Observed
Occupancy Rate	Percentage of hospital beds occupied relative to capacity	50.36%
Total Patients	Total distinct/aggregate patient count	10K
Bed Utilization Rate	Rate of bed usage relative to available beds (Resource Utilization view)	0.50
Department Efficiency	Comparative Average LOS across departments, used as an efficiency proxy	Varies by department
Available Beds	Total unoccupied beds across selected departments/hospitals	3M
ICU Beds	Total ICU bed capacity	527K
Staff	Total staff allocated across departments	3M
Occupied Beds	Total beds currently in use	5K

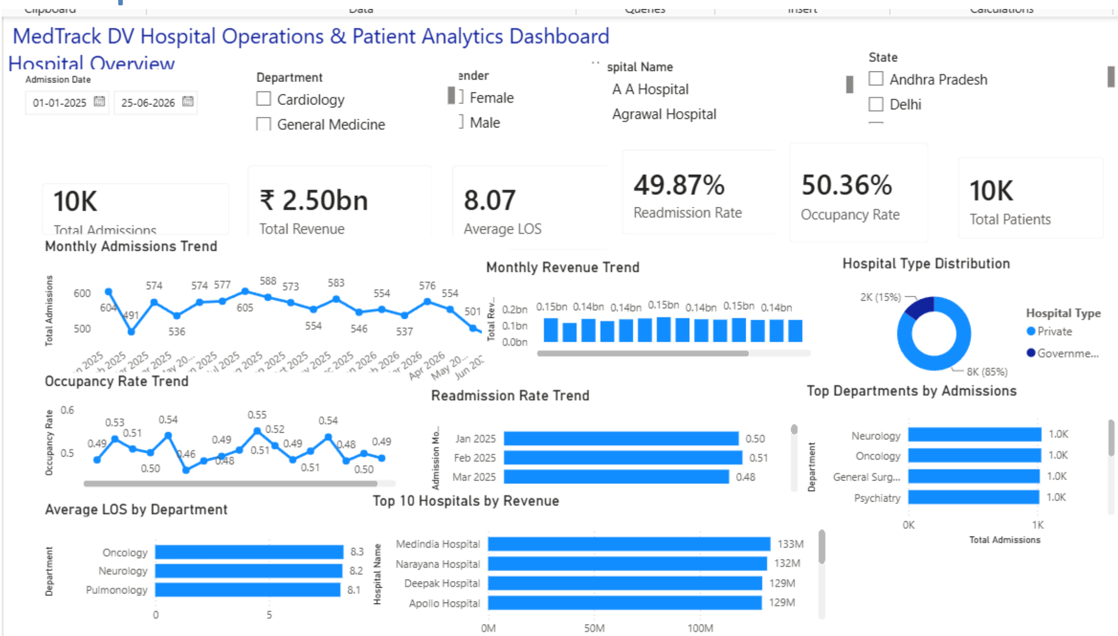
KPI values above are as displayed in the implemented dashboards at the time of the screenshots and will vary with filter selections.

13. Dashboard Architecture

```
flowchart TD
    Model[Power BI Data Model] --> D1[Hospital Overview]
    Model --> D2[Patient Flow]
    Model --> D3[Department Analytics]
    Model --> D4[Resource Utilization]
    D1 -.shared filters.- D2
    D2 -.shared filters.- D3
    D3 -.shared filters.- D4
```

All four dashboards share a common KPI card layout and a synchronized global filter panel (Admission Date, Hospital Name, Department, Gender, State), allowing stakeholders to move between operational, patient-level, department-level, and resource-level views while maintaining the same filter context.

14. Hospital Overview Dashboard



Hospital Overview Dashboard

Purpose: Provide a hospital-wide performance snapshot for administrators — combining admissions, revenue, occupancy, and readmission trends with hospital- and department-level comparisons.

KPIs Displayed: Total Admissions (10K), Total Revenue (₹2.50bn), Average LOS (8.07), Readmission Rate (49.87%), Occupancy Rate (50.36%), Total Patients (10K).

Visualizations:

Visual	Analytical Purpose
Monthly Admissions Trend (line chart)	Tracks admission volume month-over-month to identify seasonality or growth patterns
Monthly Revenue Trend (bar chart)	Shows revenue generated per month to monitor financial performance over time
Hospital Type Distribution (donut chart)	Compares admission share between Private (85%) and Government (15%) hospitals
Occupancy Rate Trend (line chart)	Tracks bed occupancy percentage over time to flag capacity pressure
Readmission Rate Trend (bar chart)	Tracks monthly readmission rate to monitor care-quality trends
Top Departments by Admissions (bar chart)	Ranks departments (Neurology, Oncology, General Surgery, Psychiatry) by patient volume
Average LOS by Department (bar chart)	Compares average length of stay across departments (Oncology, Neurology, Pulmonology)

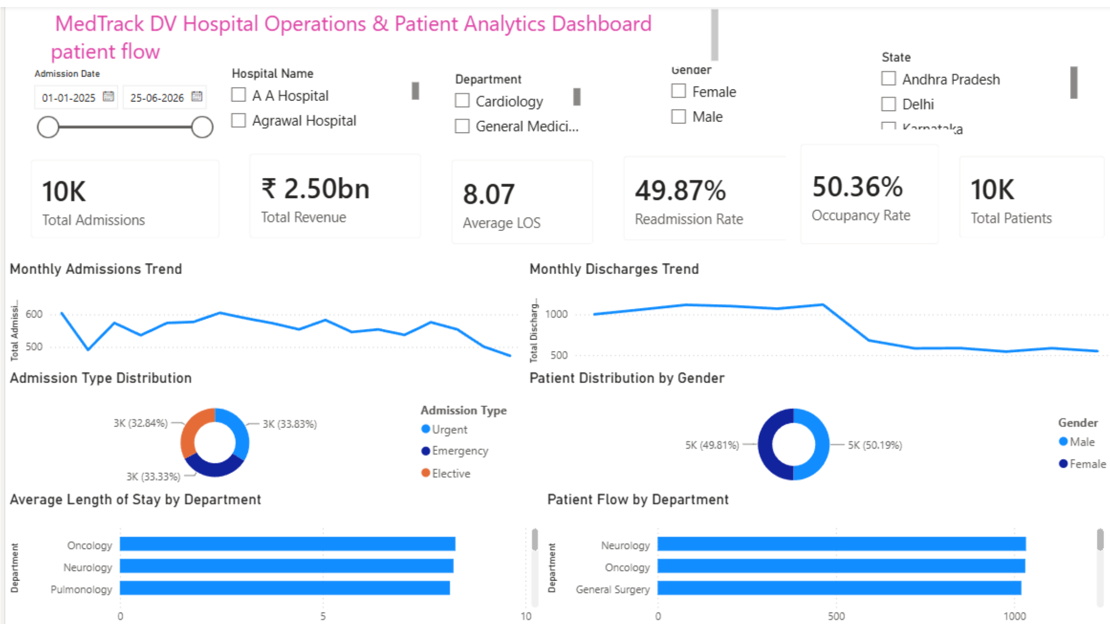
Visual	Analytical Purpose
Top 10 Hospitals by Revenue (bar chart)	Ranks hospitals (Medindia, Narayana, Deepak, Apollo, etc.) by total revenue generated

Filters: Admission Date, Department, Gender, Hospital Name, State.

What Users Can Analyze: Overall hospital performance trends, which hospital types and hospitals drive the most revenue, which departments handle the highest patient volume, and how occupancy and readmissions are trending.

Business Value: Gives leadership a single view to track financial and operational health, benchmark hospitals and departments, and detect early warning signs (rising readmissions, occupancy strain).

15. Patient Flow Dashboard



Patient Flow Dashboard

Purpose: Analyze how patients move through the hospital system — from admission to discharge — segmented by admission type, gender, and department.

KPIs Displayed: Total Admissions (10K), Total Revenue (₹2.50bn), Average LOS (8.07), Readmission Rate (49.87%), Occupancy Rate (50.36%), Total Patients (10K).

Visualizations:

Visual	Analytical Purpose
Monthly Admissions Trend (line chart)	Tracks incoming patient volume over time
Monthly Discharges Trend (line chart)	Tracks outgoing patient volume over time, enabling admission-vs-discharge comparison

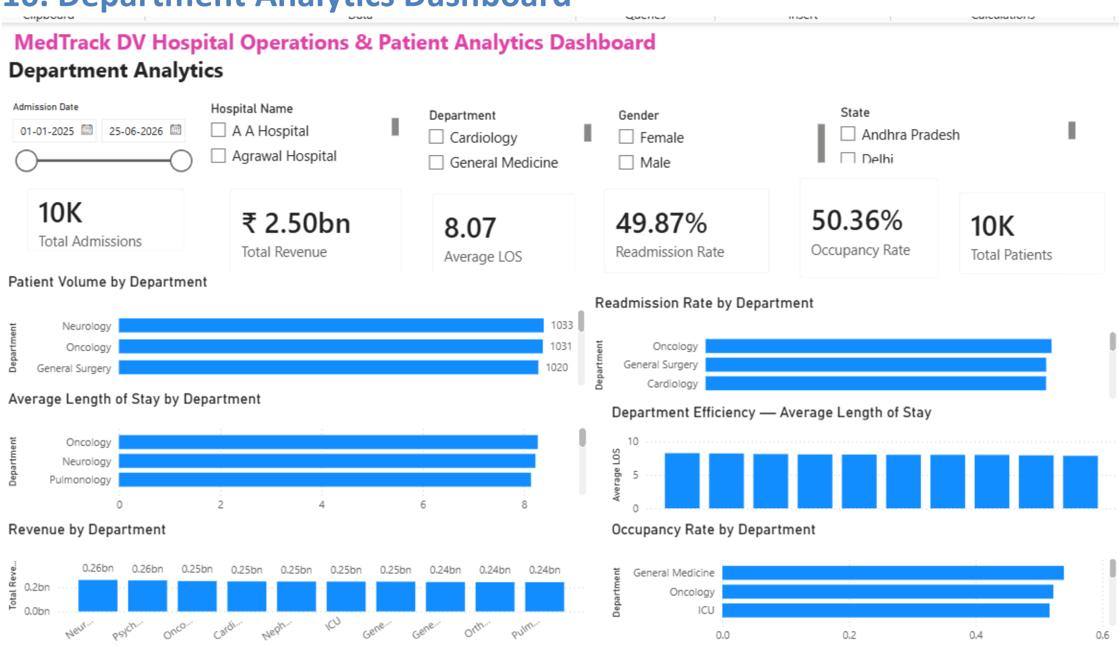
Visual	Analytical Purpose
Admission Type Distribution (donut chart)	Breaks down admissions by Urgent (32.84%), Emergency (33.83%), and Elective (33.33%)
Patient Distribution by Gender (donut chart)	Shows near-even split between Male (50.19%) and Female (49.81%) patients
Average Length of Stay by Department (bar chart)	Compares LOS across departments (Oncology, Neurology, Pulmonology)
Patient Flow by Department (bar chart)	Compares patient volume moving through each department (Neurology, Oncology, General Surgery)

Filters: Admission Date (with range slider), Hospital Name, Department, Gender, State.

What Users Can Analyze: Balance between admissions and discharges, the mix of planned vs. urgent/emergency care, demographic composition of patients, and which departments see the highest patient throughput.

Business Value: Supports staffing and bed-planning decisions by revealing patient inflow/outflow patterns and helps identify departments under sustained flow pressure.

16. Department Analytics Dashboard



Department Analytics Dashboard

Purpose: Provide a department-centric performance view — comparing patient volume, efficiency, readmissions, revenue, and occupancy across departments.

KPIs Displayed: Total Admissions (10K), Total Revenue (₹2.50bn), Average LOS (8.07), Readmission Rate (49.87%), Occupancy Rate (50.36%), Total Patients (10K).

Visualizations:

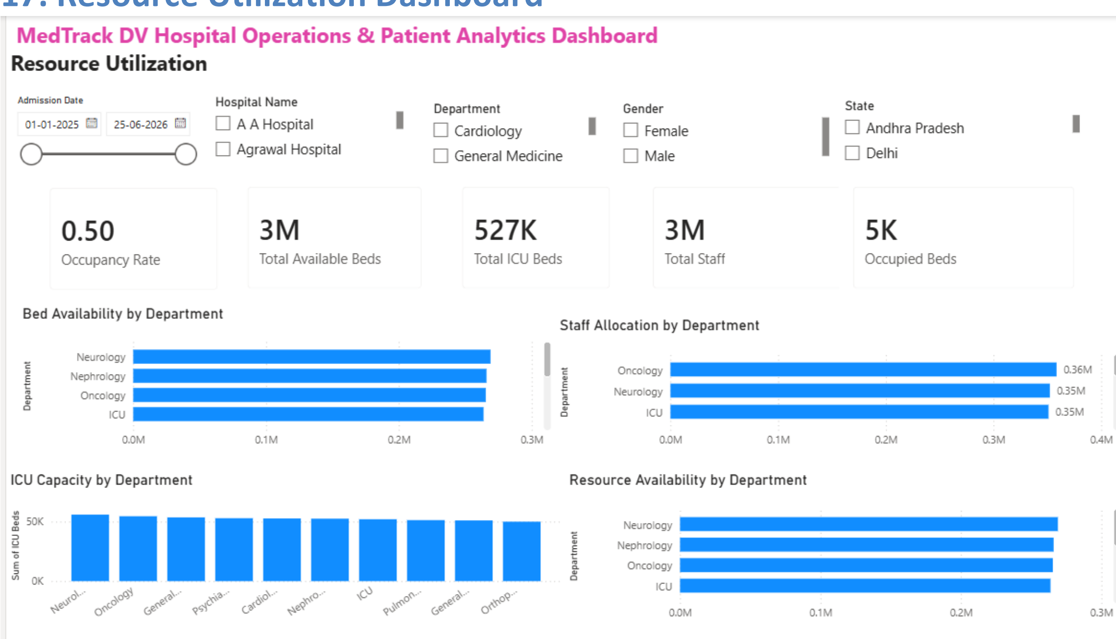
Visual	Analytical Purpose
Patient Volume by Department (bar chart)	Compares admission counts across departments (Neurology: 1,033; Oncology: 1,031; General Surgery: 1,020)
Readmission Rate by Department (bar chart)	Identifies departments with the highest readmission rates (Oncology, General Surgery, Cardiology)
Average Length of Stay by Department (bar chart)	Compares LOS across departments (Oncology, Neurology, Pulmonology)
Department Efficiency — Average LOS (bar chart)	Cross-department efficiency comparison using LOS as a proxy metric
Revenue by Department (bar chart)	Compares revenue contribution across departments (Neurology, Psychiatry, Oncology, Cardiology, Nephrology, ICU, etc., each ~0.24–0.26bn)
Occupancy Rate by Department (bar chart)	Compares bed occupancy percentage by department (General Medicine, Oncology, ICU)

Filters: Admission Date (with range slider), Hospital Name, Department, Gender, State.

What Users Can Analyze: Which departments are highest-volume vs. highest-efficiency, which departments generate the most revenue, and where readmission or occupancy risk is concentrated.

Business Value: Enables department heads and operations managers to benchmark performance across departments and prioritize quality-improvement or capacity initiatives where they matter most.

17. Resource Utilization Dashboard



Resource Utilization Dashboard

Purpose: Assess physical and staffing resource capacity — beds, ICU beds, and staff — across departments to support capacity planning.

KPIs Displayed: Occupancy Rate (0.50), Total Available Beds (3M), Total ICU Beds (527K), Total Staff (3M), Occupied Beds (5K).

Visualizations:

Visual	Analytical Purpose
Bed Availability by Department (bar chart)	Compares available bed counts by department (Neurology, Nephrology, Oncology, ICU)
Staff Allocation by Department (bar chart)	Compares staff counts allocated to each department (Oncology, Neurology, ICU)
ICU Capacity by Department (bar chart)	Compares ICU bed capacity across departments (Neurology, Oncology, General, Psychiatry, Cardiology, etc.)
Resource Availability by Department (bar chart)	Aggregated view of overall resource availability by department (Neurology, Nephrology, Oncology, ICU)

Filters: Admission Date (with range slider), Hospital Name, Department, Gender, State.

What Users Can Analyze: Which departments are resource-constrained vs. resource-rich, how staff allocation aligns with bed capacity, and where ICU capacity may need reinforcement.

Business Value: Directly supports capacity planning and staffing decisions, helping resource managers proactively rebalance beds and staff ahead of demand spikes.

18. Filters and Interactivity

All four dashboards share a consistent, synchronized global filter panel:

Filter	Type	Applies To
Admission Date	Date range picker (with slider on some pages)	All dashboards
Hospital Name	Multi-select list (e.g., A A Hospital, Agrawal Hospital)	All dashboards
Department	Multi-select list (e.g., Cardiology, General Medicine)	All dashboards
Gender	Multi-select list (Male, Female)	All dashboards
State	Multi-select list (e.g., Andhra Pradesh, Delhi, Karnataka)	All dashboards

This consistent filter design allows stakeholders to drill into a specific hospital, department, time period, or demographic segment and see the impact reflected consistently across all KPI cards and visuals on that page.

19. Business Questions Answered

- Which hospitals and departments generate the most revenue?
- How are admissions and discharges trending month over month?
- What is the current occupancy rate, and how has it trended over time?
- Which departments have the highest patient volume and the longest average length of stay?
- How are patients distributed by admission type and gender?
- Which departments have the highest readmission rates?
- How are beds, ICU capacity, and staff allocated across departments?
- Which departments are approaching resource capacity limits?

20. Key Insights

Based on the values shown in the implemented dashboards:

- **Private hospitals dominate admissions**, accounting for 85% of the Hospital Type Distribution versus 15% for Government hospitals.
- **Admission types are fairly evenly split** across Urgent (32.84%), Emergency (33.83%), and Elective (33.33%), indicating a balanced mix of planned and unplanned care.
- **Patient gender distribution is near-even** (50.19% Male / 49.81% Female).
- **Readmission Rate stands at 49.87%**, a notably high figure that department-level analysis (Oncology, General Surgery, Cardiology showing the highest rates) can help investigate further.
- **Occupancy Rate is around 50%**, suggesting moderate but consistent bed utilization across the tracked period.
- **Neurology, Oncology, and General Surgery** consistently appear as top departments by admission volume and patient flow.
- **Revenue is fairly evenly distributed across departments** (~0.24–0.26bn each), indicating no single department dominates hospital revenue generation.

(These are observational insights drawn directly from the dashboard values shown in the screenshots and should be validated against the full underlying dataset before being used for formal reporting.)

21. Data Analytics Workflow

```
flowchart LR
    A[Hospital Data] --> B[Data Collection]
    B --> C[Data Cleaning]
    C --> D[Data Transformation]
    D --> E[KPI Engineering]
    E --> F[Power BI Data Model]
```

```
    F --> G[Dashboard Development]
    G --> H[Testing & Validation]
    H --> I[Documentation]
```

22. Technology Stack

Layer	Tool / Technology
Data Cleaning & Wrangling	Python, Pandas, NumPy
Data Transformation	Power Query
Data Modeling & Calculations	Power BI, DAX (*specific DAX measures: To be confirmed*)
Visualization / Dashboarding	Power BI
Version Control & Project Management	GitHub
Documentation	Markdown

> **Platform clarification:** The original project specification proposed **Tableau** as the visualization tool. The dashboards actually implemented and documented in this repository were built in **Power BI**, as evidenced by the Power BI interface, visuals, and filter panes shown in the screenshots.

23. Project Folder Structure

```
Hospital-Operations-Patient-Analytics/
│
├── data/               # Raw and/or cleaned hospital datasets
├── scripts/            # Python scripts for data cleaning and transformation (P
andas, NumPy)
├── dashboard/           # Power BI (.pbix) dashboard file(s)
├── docs/               # Supporting documentation, reference materials, screens
hots
└── README.md           # Project documentation (this file)
```

24. Testing and Validation

The following validation practices are applicable to this type of Power BI project; specific test cases executed for this project are **To be confirmed**:

- Cross-checking KPI card totals (Total Admissions, Total Revenue, Total Patients) against source data aggregates.
- Verifying that filter interactions (Admission Date, Hospital Name, Department, Gender, State) correctly update all visuals and KPI cards.
- Confirming consistency of shared KPIs (e.g., Total Admissions, Readmission Rate) across all four dashboard pages.
- Visual inspection of chart axes, labels, and totals for accuracy against underlying values.

25. Evaluation Criteria

Consistent with standard BI project evaluation practice, this project can be assessed against:

- **Data Accuracy** — correctness of cleaned data and derived KPIs.
- **Dashboard Usability** — clarity of layout, consistent KPI placement, and intuitive filtering.
- **Analytical Depth** — relevance of visuals to real hospital operations questions.
- **Business Relevance** — how well the dashboards support real stakeholder decisions.
- **Documentation Quality** — clarity and completeness of this README and supporting docs.

(Specific weighted evaluation criteria from the original project brief were not available and are marked as To be confirmed.)

26. Project Deliverables

- Cleaned and transformed hospital dataset (Python/Power Query outputs).
- Power BI dashboard file (`.pbix`) containing all four dashboards.
- KPI definitions and data documentation.
- This README / project documentation.
- GitHub repository with organized project structure.

27. GitHub Repository

Repository: Hospital-Operations-Patient-Analytics-

Branch: `feature-sriramoji-tharun`

28. Skills Demonstrated

This project demonstrates practical, end-to-end Data Analytics / Business Intelligence capability:

Skill Area	Demonstrated Through
Data Cleaning	Handling missing/inconsistent hospital data using Python (Pandas, NumPy)
Data Transformation	Standardizing and reshaping data via Power Query
Data Modeling	Structuring cleaned data into a Power BI data model supporting cross-dashboard filtering
KPI Development	Engineering standardized metrics (Admissions, Revenue, LOS, Readmission Rate, Occupancy Rate, Utilization)
Data Visualization	Designing trend lines, bar charts, and donut charts tailored to each analytical question

Skill Area	Demonstrated Through
Interactive Dashboard Design	Building a synchronized, multi-page Power BI report with consistent global filters
Business Analysis	Translating operational data into insights relevant to hospital administrators and department heads
Healthcare Operations Analysis	Modeling hospital-specific concepts — occupancy, LOS, readmissions, ICU capacity, staffing
GitHub-Based Project Management	Structuring and version-controlling the project via a dedicated feature branch on GitHub

29. Future Improvements

- Add drill-through pages for individual hospital or department deep-dives.
- Incorporate predictive analytics (e.g., forecasting admissions or occupancy).
- Add cost-per-patient or profitability KPIs alongside revenue.
- Automate the Python data-cleaning pipeline as a scheduled refresh process.
- Expand geographic filtering with a map-based visual by state.
- Add row-level security for role-based access (e.g., department heads seeing only their department).

30. Conclusion

MedTrack DV demonstrates a complete healthcare analytics workflow — from raw data cleaning and transformation through KPI engineering to interactive Power BI dashboard delivery. By organizing hospital operations data into four focused dashboards (Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization), the project gives hospital stakeholders a consolidated, filterable view of the metrics that matter most for operational and clinical decision-making, while also serving as a practical, portfolio-ready demonstration of end-to-end data analytics and BI skills.

Repository: https://github.com/springboardmentor09876x-cmd/Hospital-Operations-Patient-Analytics-/tree/feature-sriramoji-tharun